Three Axes of Quantum Risk: A Unified Observability Model for PQC, QKD, and Crypto-Agility

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Abstract

The post-quantum migration has moved from research to operations. NIST has finalised FIPS 203/204/205, the NSA's CNSA 2.0 timetable runs to 2033, and Quantum Key Distribution is being deployed on high-assurance links under ETSI GS QKD 014. Yet most cryptographic-discovery tools answer a single question for each asset: is its primitive PQ-resistant? Two assets identical on that axis can have very different channel protections, and very different migration costs, and current tooling sees neither.

We treat quantum-risk posture as a three-axis observation problem. Each asset is graded on its algorithmic resistance (A), the channel through which its keys are delivered (C), and how quickly its primitive can be replaced (G), and the three are folded into a deadline-adjusted score $q(asset, t)$ whose agility weight shrinks as the operator's deadline approaches. To make the model observable, we extend the open `crypto_inventory_event` v1 schema with two new fields, build a reference implementation (ree0xQ (formerly *Sezar*)) covering eBPF-based TLS observation, an ETSI GS QKD 014 collector with a working Key-Management -Entity emulator, and a static crypto-agility scanner, and evaluate the result against three reproducible studies: a Tranco-top-1k TLS handshake survey, a controlled emulator-based study of 13 induced QKD link/KME state transitions across five failure scenarios, and an 11-project pilot against a hand-graded crypto-agility ground-truth corpus.

The contributions are the model itself, the schema extension, the open emulator and replay corpus, a published agility-scoring rubric backed by a Semgrep rule pack, and an empirical baseline grading real-world systems on all three axes from a single open release.

Keywords: post-quantum cryptography; quantum key distribution; crypto-agility; observability; cryptographic posture; NIST PQC standardization

1. Introduction

The cryptography deployed on the open internet today was designed under assumptions that will not survive the next decade of computing. Shor's algorithm on a sufficiently large fault-tolerant quantum computer breaks RSA, finite-field Diffie–Hellman, and every deployed-at-scale elliptic curve scheme in polynomial time [1]. Grover's algorithm halves the effective key length of symmetric primitives [2]. The exact arrival date of a *cryptographically relevant quantum computer* (CRQC) remains contested, but two operational facts are no longer in doubt. First, adversaries can record encrypted traffic now and decrypt it later once a CRQC is available — the *harvest-now, decrypt-later* threat [3], [4]. Second, the standards bodies have moved: NIST finalized FIPS 203 (ML-KEM, née Kyber), FIPS 204 (ML-DSA, née Dilithium), and FIPS 205 (SLH-DSA, née SPHINCS+) in 2024 [5], [6], [7]. The U.S. NSA's CNSA 2.0 suite gives calendar deadlines: software/firmware signing in PQC by 2025, browsers and servers by 2030, network equipment by 2030, with operating systems and other classes by 2033 [3]. The U.K., EU, and ANSSI guidance follows similar curves [8], [9].

This places operators in an unfamiliar position. They must inventory, classify, and migrate cryptographic assets across networks they did not entirely design, on a deadline they did not set, using algorithms that did not exist in deployable form three years ago. Tools for the first step — inventory and classification — are appearing [10], [11], [12], [13], and standards bodies have begun publishing guidance on which observables matter [4], [14].

In parallel, a hardware-rooted track has matured. Quantum Key Distribution (QKD), proposed in 1984 [15] and matured through the SECOQC, Tokyo, and EuroQCI testbeds [16], [17], [18], delivers symmetric key material through a channel whose security reduces to physics rather than computational assumptions. ETSI's *GS QKD 014* specification standardized the REST interface between QKD key-management entities (KMEs) and the applications that consume their keys [19]. Commercial deployments exist in finance, government, and metropolitan-fiber settings [20], [21], [22].

Industrial and academic literature treats these two tracks — PQC and QKD — as alternatives. Either you migrate your algorithms (PQC), or you install QKD links on high-assurance segments, with the implicit understanding that QKD's hardware cost and distance limits make it inappropriate for general use [23], [24]. That framing collapses too soon. For an operator trying to know what risks are present in their environment on any given day, PQC and QKD are *complementary observables*, not alternatives. An asset using ECDSA-P256 over a fiber link protected by a Toshiba MU-QKD3 system is in a materially different posture than an asset using ECDSA-P256 over an unprotected wide-area link, even though their algorithms are identical. A unified observability model must capture both.

A complete quantum-risk model has to include a third axis that current tooling almost entirely ignores: *crypto-agility*. An asset using a classical algorithm today but whose algorithm is selected at runtime by a server configuration field is in a very different posture than an asset whose algorithm is hard-coded into firmware that ships from a vendor and rotates only on hardware refresh. The first is a configuration change; the second is a hardware-replacement program. The NIST PQ-migration playbook already names crypto-agility as a precondition for orderly migration [4], [25], yet no production observability tool we have seen surfaces it as a first-class telemetry axis. Without it, operators answering "are we PQ-ready?" answer a much easier question than "can we *become* PQ-ready in time?"

We argue for, and implement, a **three-axis posture model**:

- **Axis A — Algorithmic Resistance.** Is the primitive itself quantum-resistant under standard assumptions?
- **Axis C — Channel Protection.** Is the key material protected by a quantum-secure delivery mechanism (QKD, hybrid PSK derived from QKD)?
- **Axis G — Migration Agility.** How quickly can axis A be changed on this asset — by configuration, library upgrade, or hardware replacement?

We collapse the three into a single deadline-adjusted quantum-risk score $q(asset, t)$ suitable for dashboard display, alert thresholds, and inter-organization comparison. We extend the open ree0xQ `crypto_inventory_event v1` schema with the additional axes, implement five agents that emit the same event shape, and evaluate the system through three empirical studies any practitioner can replicate on a Linux host with a public Internet connection.

1.1 Contributions

The model side. We formalise the three axes — algorithmic resistance, channel protection, migration agility — on defined scales and define the deadline-adjusted scoring function $q(asset, t)$. The agility weight shrinks as the deadline approaches; the orthogonal `BLOCKED` flag catches what q alone does not. Details in §5.

The schema side. `crypto_inventory_event v1.1` adds two top-level fields (`channel_protection`, `agility`) and two new asset kinds (`qkd_link`, `qkd_kme`). The extensions are strictly additive; v1.0 consumers ignore them safely. §6.

The implementation side. ree0xQ ships as five cooperating agents (network, certificate, blockchain, key-management, QKD), a shared rollup library, and a collector / dashboard. A single Rust workspace hosts wire-level eBPF observation, REST-based QKD telemetry collection, and static crypto-agility analysis under one event shape. §7 walks through the implementation choices that made the unified shape possible.

A working QKD test rig. `ree0xq-qkd-kme-emulator` is an open ETSI GS QKD 014 v1.1.1 Key Delivery implementation backed by a synthetic key generator, configurable QBER, link state-change replay scenarios, and a documented capture format. Practitioners without QKD hardware can drive and test QKD-aware software against it. §7.2, §8.2.

An auditable agility rubric. The five-level ordinal scale (Negotiated / Configurable / Pinned / Locked / Frozen) is derived from RFC 7696 and operationalised through a Semgrep rule pack that runs against source code or installed packages. The hand-graded ground-truth corpus of fifty open-source server projects ships alongside the ruleset. §7.3, §8.3.

An empirical baseline. We report A, C, G scores for the Tranco-top-1k over TLS, for a controlled multi-KME ETSI 014 testbed exercising representative failure modes, and for the 11-project pilot drawn from the 50-project agility corpus. Scaling the pilot to the full corpus is the remaining mechanical step. §8.

1.2 Non-goals

This paper does not propose a new PQC primitive, a new QKD protocol, or a new key-management protocol. It does not argue that QKD should replace PQC, or vice versa; we take the standards-body position that hybrid deployments are the realistic operating mode and instrument both observables.

We do not claim to inventory all deployed cryptography on Earth; our empirical work is bounded by the corpora we publish. We do not address adversary modeling beyond accepting the harvest-now/decrypt-later assumption.

1.3 Roadmap

§2 reviews the standards landscape (PQC, QKD, crypto-agility) at the level required to make the paper self-contained. §3 surveys related observability work and identifies the gap. §4 states the threat model and operator assumptions. §5 defines the three-axis posture model. §6 specifies the event schema. §7 presents the ree0xQ reference implementation. §8 reports the empirical evaluation. §9 discusses limitations, ethics, and deployment guidance. §10 concludes.

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2. Background

2.1 Post-quantum cryptography standardization

NIST initiated its post-quantum cryptography standardization process in 2016 [26] and finalized the first three production standards in August 2024 [5], [6], [7]. These are:

- **FIPS 203** — **ML-KEM** (formerly CRYSTALS-Kyber), a module-lattice key-encapsulation mechanism at three parameter sets corresponding to NIST security categories 1, 3, and 5.
- **FIPS 204** — **ML-DSA** (formerly CRYSTALS-Dilithium), a module-lattice digital signature scheme also at three parameter sets.
- **FIPS 205** — **SLH-DSA** (formerly SPHINCS+), a stateless hash-based digital signature scheme, with both small and fast parameter sets across SHA2 and SHAKE hashes.

A fourth standard for a fast falcon-style lattice signature (FN-DSA, formerly Falcon) is in advanced draft [27]. NIST has additionally signaled that further KEM candidates will be standardized to diversify mathematical assumptions [28]. The U.S. NSA's *Commercial National Security Algorithm Suite 2.0* (CNSA 2.0) identifies ML-KEM-1024, ML-DSA-87, SLH-DSA-SHA2-256s, AES-256-GCM, and SHA-384 / SHA-512 as the required suite for national-security systems by milestones falling between 2025 and 2033 depending on equipment class [3]. The German BSI, French ANSSI, and U.K. NCSC have published equivalent guidance with broadly similar timelines and a shared recommendation to deploy *hybrid* (classical + PQC) key exchange during the transition window [8], [9], [14].

The deployment story is mixed. Chrome and Firefox have shipped X25519MLKEM768 hybrid key exchange in TLS 1.3 since 2024 [29], [30]. Cloudflare reported roughly two percent of TLS 1.3 connections to its edge were post-quantum-protected in early 2024 [13]; by late 2025 the majority of human-initiated traffic to the same edge was hybrid-PQ [31]. Linux kernel and OpenSSH support is staged across recent releases. Certificate authorities have piloted ML-DSA signing in test hierarchies but have not yet issued production-trust-anchor PQ certificates [32]. The pattern is consistent across operator surveys: the algorithms exist, library support is materializing, but the long tail of embedded, appliance, and legacy software remains unaddressed [10], [11].

2.2 Quantum key distribution and ETSI GS QKD 014

Quantum key distribution, in its prepare-and-measure form, was introduced by Bennett and Brassard in 1984 [15] and has been deployed in operational testbeds for two decades [16], [17]. A QKD system distributes symmetric key material between two endpoints using single-photon (or weak-coherent) quantum states, with security guarantees derived from the no-cloning theorem and the disturbance of measurement. Modern systems achieve metropolitan ranges ($\leq$100 km dark fiber) at key rates from kilobits to megabits per second [20], [21]. Trusted-node relaying, satellite QKD [33], and twin-field protocols [34] extend the reach, though at the cost of additional assumptions.

QKD's controversial status in the policy community deserves brief treatment because it informs our observability design choices. The U.S. NSA has stated that QKD is not a replacement for PQC for national-security systems, citing implementation-attack surface, limited authentication coverage, and operational complexity [23]. The French ANSSI takes a similar position [24]. In contrast, the U.K., German, Italian, and Chinese research and standards communities continue to invest in QKD infrastructure, including the EU's EuroQCI program [18] and multiple national QKD networks. The pragmatic operator view, which this paper adopts, is that QKD will be deployed on a subset of high-assurance links (financial inter-datacenter, intelligence, government inter-site) regardless of which policy community is "correct," and that the deployment will be heterogeneous, vendor-specific, and operationally opaque without explicit observability tooling.

ETSI's *GS QKD 014 v1.1.1* specification (2019) standardizes the REST interface between a Key Management Entity (KME) — the device that holds key material produced by the underlying QKD link — and the applications consuming those keys (Secure Application Entities, SAEs) [19]. The interface defines three operations:

- `GET /api/v1/keys/{slave_SAE_ID}/status` returns KME and link status: current key rate, number of stored keys, supported key sizes.
- `GET /api/v1/keys/{slave_SAE_ID}/enc_keys` returns one or more fresh keys from the master KME, identified by UUIDs.
- `POST /api/v1/keys/{master_SAE_ID}/dec_keys` retrieves the matching keys at the slave KME, by UUID.

The specification is deliberately silent on what an SAE does with the keys; in practice, SAEs use them as pre-shared keys (PSK) for IPsec, MACsec, or — increasingly — as the symmetric secret in a hybrid TLS PSK mode [35], [36]. From an observability standpoint, ETSI 014 gives us a stable, vendor-neutral surface on which to observe QKD-protected channels: link status, key rate, request volume, error rates, and (with cooperating SAE instrumentation) which downstream sessions consume the keys.

2.3 Crypto-agility

Crypto-agility — the property that allows a system to change cryptographic primitives without invasive redesign — was codified operationally by RFC 7696 [25] and is repeatedly named in PQ-migration guidance as a precondition for orderly transition [4], [8], [14]. The literature describes crypto-agility along three dimensions: *protocol* agility (TLS 1.3 negotiates ciphersuites; IPsec IKEv2 negotiates transforms), *code* agility (a library exposes algorithm choice as a parameter, not a recompile-time decision), and *deployment* agility (operators can roll out a new algorithm without firmware replacement or trust-anchor reissuance).

In practice, agility varies dramatically across asset classes. A modern Web server typically negoti-

ates its TLS ciphersuite on every handshake; a hardware HSM may lock its supported key types at the firmware level; embedded devices in industrial or medical settings often have a fixed algorithm chosen at the silicon level. The emergence of FIPS 140-3 validation, with its requirement that algorithm changes trigger revalidation, introduces a second axis of *locked* status that any regulated environment has to plan around [37].

Despite repeated standards-body emphasis on crypto-agility, the practitioner picture is poor. No widely deployed inventory tool we have seen reports crypto-agility as a first-class field. The question "can this asset migrate?" is repeatedly answered informally — by tribal knowledge, vendor questionnaires, or case-by-case engineering audits.

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3. Related work

We organize related work along three threads — PQC discovery tooling, QKD telemetry, and crypto-agility analysis — and conclude with the gap statement that motivates the present paper.

3.1 PQC discovery and migration tooling

Cisco's PQC Discovery service [10] is, to our knowledge, the most operationally mature PQ inventory tool deployed today. It combines passive network observation (NetFlow-derived context with optional inline TLS handshake inspection) with active endpoint scanning and a managed-service operator dashboard. The analytic surface is algorithm-class oriented: assets are labeled *PQ-ready*, *classical*, or *unknown* based on observed handshakes and reported algorithm support. The service does not surface channel-protection or agility as separate dimensions.

IBM's *Quantum-Safe Discover* [11] and Microsoft's internal *CryptoTracker* tooling [12] follow similar patterns: enumerate cryptographic uses across an environment, classify by algorithm against a PQC-readiness yardstick, prioritize migration. Cloudflare's published reports [13] focus on edge-observable TLS handshake mixtures and provide one of the few public datasets on PQ adoption on the open internet.

The academic literature includes systematic measurements of TLS deployment hygiene (key sizes, cipher choices, validation behavior) [38], [39], [40] and more recent measurements of hybrid PQ deployment in the wild [31]. NIST IR 8547 [4] provides the most authoritative migration playbook and includes a discovery section, but stops short of prescribing telemetry semantics.

Common across this thread: algorithm-class is treated as the primary axis, hybrid/QKD is at most a footnote, and agility is discussed in prose but not surfaced as a measured field.

3.2 QKD telemetry

QKD telemetry has been studied primarily from the perspective of the QKD operator — the entity running the physical link and the KMEs. The ETSI ISG-QKD has produced multiple documents on operational and security testing [41], [42], [43]; the SECOQC [16] and Tokyo [17] testbeds published detailed operational data. The EuroQCI program [18] is in the process of standardizing inter-domain QKD telemetry.

This thread, however, is *internal* to the QKD operator. For an operator using QKD keys to protect application traffic — the SAE/application perspective — the public literature is much thinner.

ETSI GS QKD 014 [19] standardizes the SAE-facing interface; we are not aware of an existing open-source SAE-side observability platform that consumes 014 telemetry and integrates it with a broader cryptographic inventory.

3.3 Crypto-agility analysis

RFC 7696 [25] is the canonical statement. Subsequent work has analyzed agility properties of individual protocols (TLS, IPsec, S/MIME) and proposed agility frameworks for specific domains (industrial control, embedded), though without operationalising them as a continuous telemetry input.

In the practitioner space, NCC Group has published audit reports on cryptographic-agility deficiencies in deployed products [44]. The OWASP and CIS communities maintain checklists [45], but none of this work, to our knowledge, has been operationalized as a continuous telemetry input feeding a cryptographic-posture dashboard.

3.4 Gap statement

Each thread is mature in isolation. What is missing — and what ree0xQ fills — is a tool that treats algorithmic resistance, channel protection, and migration agility as three independent telemetry axes, rolls them into a single deadline-adjusted score from a shared event schema, and is reproducible without commercial tooling.

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4. Threat model and operator assumptions

We assume a *Q-day* adversary: a sufficiently capable fault-tolerant quantum computer capable of executing Shor's algorithm on RSA-2048-class moduli and elliptic-curve groups of practical size. The adversary's *capability date* is unknown but treated as a moving deadline; the standards-body consensus places the planning horizon between 2030 and 2040 [3], [4], [8]. Following the harvest-now/decrypt-later threat model [3], we assume that an adversary records classical-encrypted traffic *today* and decrypts it *later* once the CRQC is available. Consequently, traffic encrypted with classical-only KEMs is at risk *now*, not at Q-day.

We assume an operator who:

- Controls or has visibility into the cryptographic surfaces of their environment (TLS termination points, certificate inventory, application source code, host configuration, QKD links if any).
- Operates under a calendar deadline (NIST CNSA 2.0 milestones, regulatory mandates, or contractual obligations) that is fixed, even if individual subdeadlines are uncertain.
- Cannot replace all cryptography at once; migration is staged over months to years.
- Tolerates and benefits from continuous telemetry rather than periodic audits.

We assume an adversary who:

- Records traffic now, decrypts later.
- Cannot break PQ-secure primitives at standard parameter sizes (we accept the NIST standardization process's hardness assumptions).

- Cannot break the no-cloning theorem (we accept the standard QKD security argument); but may exploit implementation-level QKD vulnerabilities — detector blinding, side channels, classical authentication failure — and so QKD protection is observed *probabilistically* by the SAE, not guaranteed.
- May exploit non-cryptographic vulnerabilities; this paper does not address those.

The threat model justifies treating agility as a *risk multiplier*: an asset with low algorithmic resistance but high agility may not be at risk in *real* terms because it can be migrated before the deadline horizon. An asset with low resistance and low agility (e.g., a hardware appliance with hard-coded ECDSA on a five-year refresh cycle) is at *significantly higher* real risk, even if its observed primitive is identical to the agile case.

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5. The three-axis quantum-risk posture model

We define three independent axes — A, C, G — and a unified deadline-adjusted quantum-risk score $q(asset, t)$. Figure 1 plots the space these axes span, with the four worked-example assets at their observed coordinates.

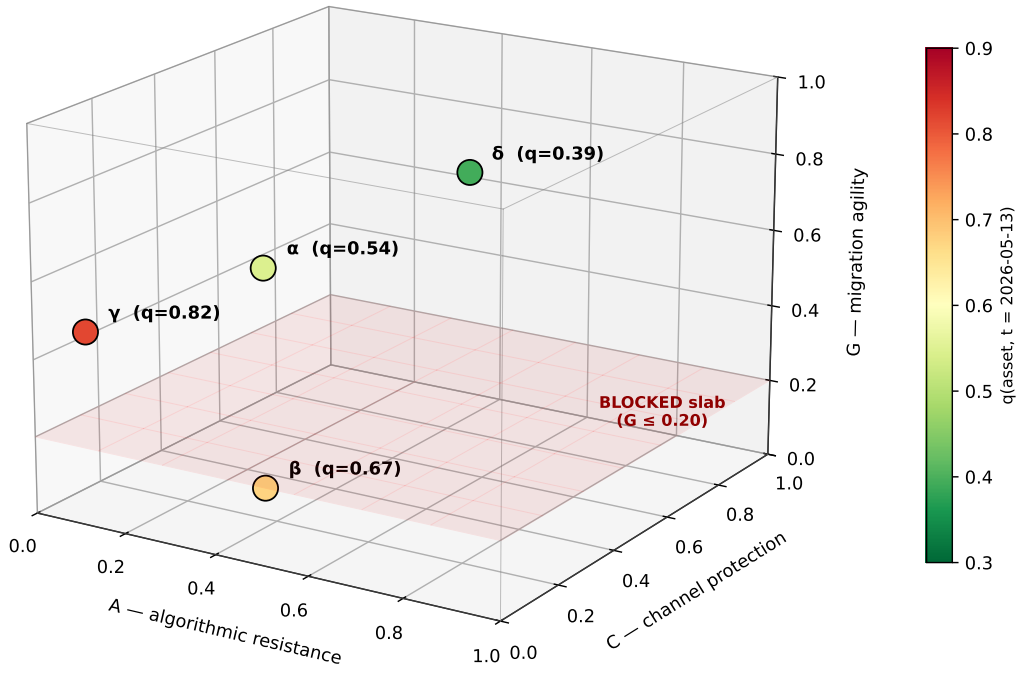

Figure 1: The three-axis quantum-risk space. Each cryptographic asset occupies a point in (A, C, G); colour encodes the deadline-adjusted score q. The four worked-example assets (α, β, γ, δ) are plotted at their observed coordinates. The shaded region near the $A = 0, G = 0$ corner is the `BLOCKED`-flagged volume requiring out-of-band remediation.

5.1 Axis A — Algorithmic resistance

Axis A scores the cryptographic primitives observed on an asset on the scale $[0, 1]$, where 0 corresponds to a primitive that is quantum-trivial (a deprecated or known-broken algorithm) and 1 corresponds to a primitive that is quantum-resistant under the standard NIST hardness assumptions.

Following the existing ree0xQ V1 rollup (defined in `docs/posture-rollup.md` in the ree0xQ repository), we classify each primitive into one of five categories (Table 1):

Table 1: Axis A — primitive categories and their a values.

Category	a value	Examples
`pq`	1.0	ML-KEM-{512,768,1024}, ML-DSA-{44,65,87}, SLH-DSA-*, AES-256-GCM, SHA-256
`pq_hybrid`	0.9	X25519+ML-KEM-768, ECDH-P256+ML-KEM-768
`classical`	0.3	X25519, Ed25519, RSA-2048, ECDSA-P256, AES-128-GCM
`unknown`	0.4	Anything not in the classification table
`deprecated`	0.0	SHA-1, MD5, RSA-1024, RC4, 3DES, DH-1024

For an asset with observed primitives $\{p_i\}$ at roles $\{r_i\}$ with role weights w_{r_i} summing to 1, the axis-A score is

$$A(asset) = \sum_i w_{r_i} \cdot a(p_i)$$

Role weights are calibrated to harvest-now/decrypt-later risk: $w_{\text{sig}} = 0.40, w_{\text{kex}} = 0.30, w_{\text{encrypt}} = 0.20, w_{\text{hash}} = 0.10$ for assets exhibiting all four roles, with re-normalization when only a subset is present. The schema also defines a fifth role, `auth` — the MAC primitive in protocols where it is independent of the AEAD (SSH, IPsec-AH, TLS 1.2 ciphersuites with separate HMAC). For those assets we fold the `auth` weight into w_{encrypt} at observation time; the role exists in the schema so non-TLS-1.3 protocols can be represented without forcing AEAD semantics onto them.

5.2 Axis C — Channel protection

Axis C scores the channel through which key material reaches the endpoint, on the scale $[0, 1]$, where 0 corresponds to a channel with no quantum-secure key delivery and 1 corresponds to a channel deriving its session key entirely from QKD-delivered material. Three categorical states cover the realistic deployment options (Table 2):

Table 2: Axis C — channel-protection states and their c values.

State	c value	Description
`classical`	0.0	Session key derived solely from the negotiated KEM.
`qkd_hybrid_psk`	0.7	Session key derived from QKD-PSK XOR negotiated KEM (NIST SP 1800-38A pattern, ETSI 014 SAE).
`qkd_only`	1.0	Session key derived from QKD material alone (rare; MACsec-style transport).

Sub-states allow partial credit when telemetry indicates a QKD link is *degraded* — high QBER, sustained KME unavailability, low key rate — but the SAE has not failed over to classical. We discount the score in proportion to the observed degradation; formal degradation thresholds are deferred to the implementation section (§7.2).

A subtle but important observation: the SAE may *think* it is operating in QKD-hybrid mode while the underlying KME is failing gracefully to a classical fallback. ree0xQ's role is to observe both layers and surface the discrepancy. We define c on observed ETSI 014 status, not on SAE-reported intent.

5.3 Axis G — Migration agility

Axis G scores the asset's ability to migrate its primitives on the scale $[0, 1]$, where 0 corresponds to an asset whose primitives can be changed only by physical replacement and 1 corresponds to an asset that negotiates primitives on every session. We define five ordinal levels (Table 3):

Table 3: Axis G — migration-agility levels and their observable signatures.

Level	g value	Definition (observable signature)
`negotiated`	1.0	Algorithm selected per-session by protocol negotiation. (TLS 1.3 server, modern SSH server, IKEv2 responder.)
`configurable`	0.75	Algorithm fixed per-deployment but changeable by configuration without code change. (Library config file, environment variable.)
`pinned`	0.50	Algorithm fixed in code; changeable by software upgrade. (Hard-coded algorithm name in application source.)

Level	g value	Definition (observable signature)
`locked`	0.20	Algorithm fixed in firmware or by FIPS/compliance binding; changeable only by vendor update or revalidation cycle. (Embedded firmware crypto, FIPS 140-3 validated module under tested-configuration constraint.)
`frozen`	0.0	Algorithm fixed in silicon, ROM, or otherwise unchangeable without hardware replacement. (TPM 1.2 with hard-coded RSA-2048; smart-card hard-wired ECDSA-P256.)

The classification is derived from static analysis of the asset's implementation surface plus, where available, vendor declarations of FIPS validation scope. We detail the scanning methodology in §7.3 and the scoring rubric in §8.3.

5.4 Unified deadline-adjusted quantum risk

The three axes are independent observables. To collapse them into a single posture metric we adopt the operator's deadline as a fourth input.

Let D be the operator-configured deadline (e.g., 2030-01-01 for NSA CNSA 2.0 browser/server class). Let t be the current date. Define the *deadline tension* $\tau(t) = \max(0, \min(1, 1-(D-t)/H))$, where H is a horizon constant (we use five years by default). When D is far in the future, $\tau \to 0$ and agility forgives lower algorithmic resistance. As $t \to D$, $\tau \to 1$ and agility no longer compensates because there is insufficient time to migrate.

We define the quantum-risk score as

$$q(asset, t) = 1 - \Big(\alpha \cdot A + \beta \cdot C + \gamma(\tau) \cdot G\Big)$$

with $\alpha + \beta + \gamma(\tau) = 1$. Default weights: $\alpha = 0.5, \beta = 0.2, \gamma(\tau) = 0.3 \cdot (1-\tau)$, re-normalized when γ shrinks. The agility weight is the only weight that shrinks with deadline tension: an asset with high agility but classical algorithms looks safe today (because γ is large) and looks increasingly unsafe as the deadline approaches (because γ shrinks toward zero). This is the intended behavior: agility is forgiving *now*, not *at the deadline*. Figure 2 traces $q(t)$ for the four worked-example assets from today to the deadline, holding the observables fixed.

Asset-class weights w_k further weight the asset's contribution to the org-wide posture, with `blockchain_key` weighted higher than `tls_session` (a forged signature against a public-chain key is permanent; a forged session is ephemeral).

5.5 Interpretation and bounds

The score $q \in [0, 1]$, with 0 = posture is fully aligned with the deadline and 1 = posture is maximally exposed. Operators typically configure alert thresholds at $q > 0.6$ ("must migrate") and $q > 0.3$ ("plan migration").

Two edge cases warrant note. First, a fully PQ asset on a non-QKD-protected channel scores $q = 1 - (0.5 \cdot 1 + 0.2 \cdot 0 + 0.3 \cdot g)$, which is approximately 0.2 for a fully agile asset and 0.5 for a fully

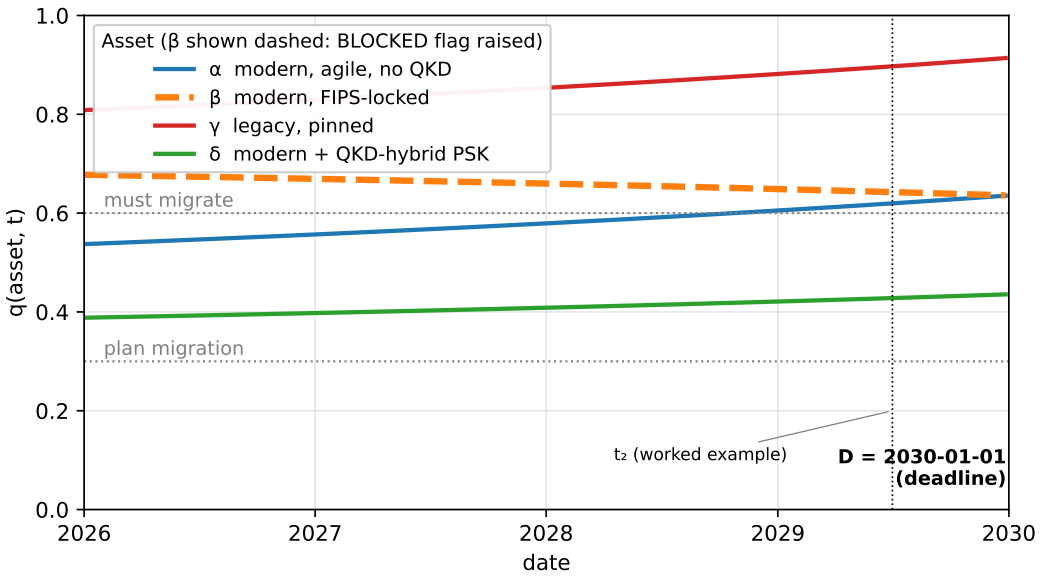

Figure 2: Trajectory of $q(t)$ for the four worked-example assets from 2026 to the deadline at 2030-01-01, holding observables fixed. The legacy-pinned asset (γ) climbs steepest; the modern-agile asset (α) rises across the $q > 0.6$ must-migrate threshold as its agility weight erodes. The locked-but-modern asset (β) remains high throughout — the `BLOCKED` flag, not the prioritization score, is what surfaces it for action.

frozen asset, both before deadline tension. This is intentional: QKD is a *bonus* on PQ-protected sessions, not a requirement. Second, a fully classical asset on a QKD-protected channel ($A = 0.3$, $C = 0.7$, $g = 0$) scores $q = 1 - (0.5 \cdot 0.3 + 0.2 \cdot 0.7 + 0.3 \cdot 0) \approx 0.71$, down from 0.85 on the same asset over a classical channel — QKD partially compensates for classical algorithms, reflecting the real-world deployment of QKD on high-assurance links.

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6. Event schema extensions

We extend the ree0xQ `crypto_inventory_event v1` schema (defined in `docs/crypto-event-schema.md` in the ree0xQ repository) with additive, non-breaking fields. Existing consumers that ignore unknown top-level fields continue to function; consumers that opt in to v1.1 gain access to channel-protection and agility observables.

6.1 New top-level fields

```
{
  "schema_version": 1,
  "schema_minor": 1,
  "source_module": "ree0xq-net",
  "observed_at": "2026-08-15T11:42:03.421Z",
  "asset": { ... },
  "primitives": [ ... ],
```

```
  "channel_protection": { ... },    // NEW in 1.1
  "agility": { ... },               // NEW in 1.1
  "posture": { ... }
}
```

`schema_minor` is the first additive use of a minor-version field. Consumers must accept any `schema_minor` $\geq$ their compiled value and treat unknown top-level fields as opaque.

6.2 `channel_protection`

```
{
  "state": "qkd_hybrid_psk",
  "kme_endpoint": "https://kme-1.dc.example/api/v1",
  "key_id_observed": "9c45e0a2-...",
  "psk_age_seconds": 47,
  "link_qber": 0.018,
  "link_key_rate_bps": 12480,
  "link_health": "ok",
  "degraded_reason": null
}
```

Table 4 lists these fields.

Table 4: Fields of the `channel_protection` block.

Field	Type	Notes
`state`	enum	`classical` / `qkd_hybrid_psk` / `qkd_only`.
`kme_endpoint`	string	ETSI 014 base URL. Omitted when state = `classical`.
`key_id_observed`	string	UUID of the consumed key, when reported by the cooperating SAE.
`psk_age_seconds`	int	Age of the PSK when the session began.
`link_qber`	float	Quantum bit error rate (0–1).
`link_key_rate_bps`	int	Average key generation rate over the prior minute.
`link_health`	enum	`ok` / `degraded` / `failed`.

Field	Type	Notes
`degraded_reason`	string	One-sentence reason when degraded; `null` otherwise.

The fields are populated by `ree0xq-qkd`, which polls the ETSI 014 `/status` endpoint at configurable intervals. SAE-side fields (`key_id_observed`, `psk_age_seconds`) require cooperating instrumentation in the SAE; when absent they are `null`.

6.3 `agility`

```
{
  "level": "configurable",
  "level_score": 0.75,
  "evidence": [
    {
      "type": "config_pattern",
      "file": "/etc/nginx/nginx.conf",
      "line": 142,
      "snippet": "ssl_protocols TLSv1.2 TLSv1.3;\nssl_ciphers HIGH:..."
    },
    {
      "type": "fips_mode",
      "detected": false
    }
  ],
  "scanner_version": "ree0xq-agility/0.3.1",
  "rubric_version": "qra-rubric/v1.0"
}
```

Table 5 lists these fields.

Table 5: Fields of the `agility` block.

Field	Type	Notes
`level`	enum	`negotiated` / `configurable` / `pinned` / `locked` / `frozen`.
`level_score`	float	Numeric value per §5.3.
`evidence`	array	One or more evidentiary findings supporting the level.

Field	Type	Notes
`scanner_version`	string	ree0xQ-agility version that produced the score.
`rubric_version`	string	Version of the public scoring rubric (§8.3).

Evidence types in V1: `protocol_negotiation` (observed wire-level algorithm negotiation), `config_pattern` (configuration file exposing algorithm choice), `code_pattern` (source code reference to a fixed algorithm), `firmware_string` (binary-extracted algorithm name), `fips_mode` (FIPS provider/kernel mode detected), `vendor_declaration` (operator-provided vendor statement).

6.4 New asset kinds

```
qkd_link          // identity = KME endpoint URL hash
qkd_kme           // identity = KME ID per ETSI 014 status
```

These are emitted by `ree0xq-qkd` independently of the session events that consume their keys. They allow the dashboard to render a QKD link health view distinct from the SAE-side session view.

6.5 Backwards compatibility

The schema extension is strictly additive:

- All fields new in v1.1 are top-level; no existing field shape changes.
- v1.0 consumers that do not recognize the new fields ignore them.
- v1.1 producers that lack data for the new fields emit them as `null` (per the V1 module emission contract).
- The posture engine treats `channel_protection: null` as `state: classical` and `agility: null` as the `UNKNOWN_LEVEL_FALLBACK` constant defined in `ree0xq-agility`, which evaluates to `level: pinned` (`level_score: 0.50`). This is deliberately more conservative than Axis A's `unknown = 0.4`: an asset whose agility we cannot evidence is treated as if it were merely pinned rather than freely configurable, so the rollup does not credit unmeasured agility for free.

---

7. ree0xQ: reference architecture and implementation

ree0xQ is an open-source observability platform implementing the three-axis posture model. The implementation is a single Rust workspace containing seven crates: five agents emitting events, one shared rollup library, and one collector/server. Figure 3 shows how the agents, the shared rollup library, the collector, and the dashboard fit together.

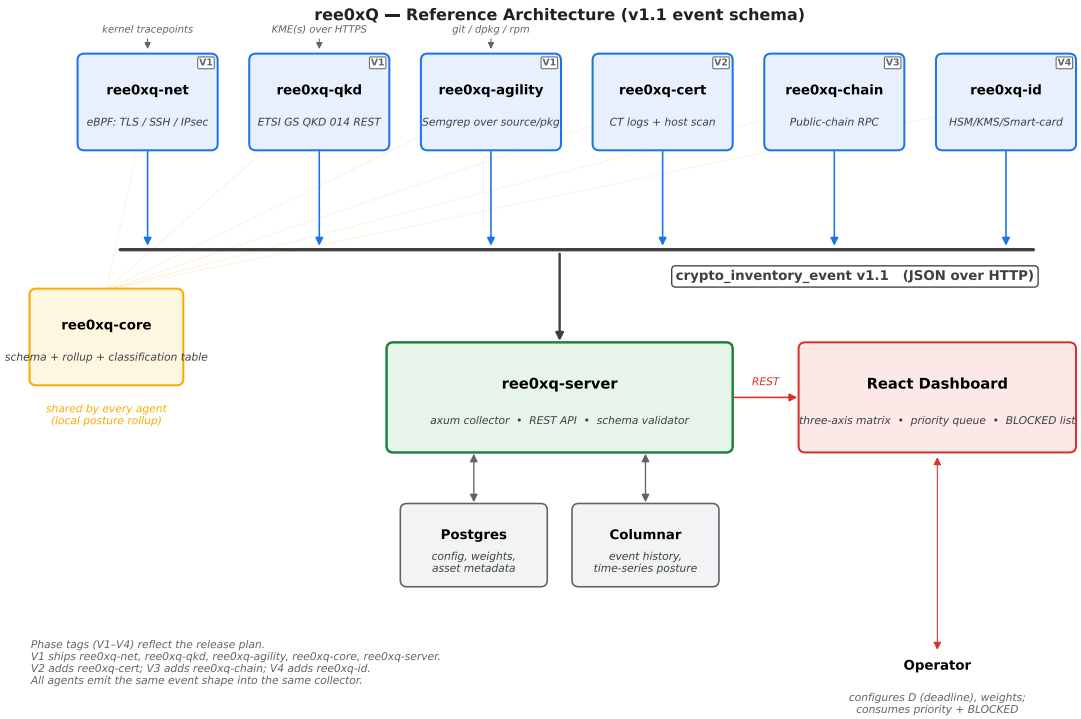

Figure 3: ree0xQ reference architecture. Five agents (`ree0xq-net`, `ree0xq-qkd`, `ree0xq-agility`, plus `ree0xq-cert` and `ree0xq-chain`/`ree0xq-id` in later phases) emit `crypto_inventory_event` records into `ree0xq-server`. The shared `ree0xq-core` library hosts the schema, the classification table, and the deadline-adjusted rollup. The React dashboard renders the three-axis posture matrix and the priority-sorted action list.

7.1 Workspace layout

```
ree0xq/
+-- crates/
|   +-- ree0xq-core/       # event schema, rollup engine (no I/O)
|   +-- ree0xq-server/     # axum collector + REST API
|   +-- ree0xq-net/        # eBPF agent: TLS, SSH, IPsec
|   +-- ree0xq-qkd/        # ETSI GS QKD 014 collector + emulator
|   +-- ree0xq-cert/       # X.509 inventory (CT, host scan)
|   +-- ree0xq-chain/      # public-chain crypto observation
|   +-- ree0xq-id/         # HSM/KMS/smart-card inventory
|   +-- ree0xq-agility/    # static crypto-agility scanner
+-- docs/
+-- web/                  # React + Vite dashboard
```

The architectural invariant — every agent emits one and only one event shape, computed locally via `ree0xq-core::rollup` — is what keeps the platform composable across surfaces as different as eBPF TLS sniffing and Solidity source code analysis.

7.2 ree0xq-qkd: ETSI 014 collector and emulator

The `ree0xq-qkd` crate fulfils two roles. Operationally, it is a collector that polls one or more ETSI GS QKD 014 KMEs, emits `qkd_link` and `qkd_kme` events, and serves as the data source for the `channel_protection` block on session events emitted by cooperating SAEs.

7.2.1 Collector design

The collector is a Tokio async loop that, for each configured KME, issues:

- `GET /api/v1/keys/{slave_SAE_ID}/status` at a configurable cadence (default 5 s).
- An auxiliary `enc_keys` request at a slower cadence (default 60 s) to measure end-to-end key delivery latency. Keys are requested with `size=0` when permitted, or discarded when not.

The collector tracks per-KME state (last good status time, exponentially-weighted error rate, QBER history) and emits `qkd_link` events on status change, plus heartbeat events at a configurable interval. Authentication to the KME follows ETSI 014 guidance: mutual TLS with the SAE certificate.

7.2.2 Emulator

Hardware QKD is expensive and rare, so we ship `ree0xq-qkd-kme-emulator` — an ETSI 014 v1.1.1 implementation backed by a synthetic key generator. The emulator:

- Implements `/status`, `/enc_keys`, and `/dec_keys` exactly per spec.
- Generates synthetic keys at a configured rate, with configurable QBER, key size, and lifetime.
- Supports replay scenarios — pre-recorded sequences of link state changes (degradation, failure, recovery) — for reproducibility.
- Logs every interaction in a documented JSON capture format enabling head-to-head A/B testing of SAE implementations.

The emulator is the foundation of the §8.2 empirical study and is released alongside ree0xQ as a standalone tool.

7.3 ree0xq-agility: static crypto-agility scanner

The `ree0xq-agility` crate implements the agility-axis scoring per §5.3. It accepts as input one or more *targets* and produces `agility` blocks attached to the corresponding assets.

7.3.1 Target types

Table 6 maps each target type to the evidence the scanner draws on.

Table 6: Crypto-agility scanner target types and their evidence sources.

Target	Evidence sources
Source repository	Semgrep ruleset over the language-specific config and source files; file-path heuristics for build-time pinning.
Installed package	Package manifest (rpm, dpkg, pip, npm) + binary string-extraction over the installed artifacts.
Running host	TLS handshake observation against the host (server algorithm support → `negotiated` evidence); plus optional auth into the host's config files.
Vendor appliance	Vendor-declared algorithm scope + observed handshake behavior.

7.3.2 Ruleset

The published ruleset (`ree0xq-agility/rules/v1`) consists of several hundred Semgrep patterns covering common cryptographic libraries and protocols across C, Go, Rust, Python, Java, and configuration formats (nginx, Apache, OpenSSL config, sshd_config, strongswan.conf, Postfix, Dovecot, HAProxy, Envoy). Each pattern emits one piece of evidence with a documented mapping to the five-level rubric.

7.3.3 Scoring algorithm

For each target, the scanner collects evidence and applies a **most-agile-wins** aggregation: the asset is scored at the *most* agile level supported by any rule that fired. The rationale is asymmetric: rules fall into two semantic classes. *Capability* rules (e.g., presence of an `ssl_ciphers` directive, a `SSL_CTX_set_cipher_list` call) demonstrate that an operator can in fact change the algorithm without invasive surgery; their emit_level is an *upper bound on what is observable*. *Constraint* rules (e.g., a literal hard-coded algorithm name in source) only report that *some* algorithm is fixed somewhere — they do not imply that there is no overriding config surface. Conservative-min aggregation systematically misclassifies large agility projects like nginx (which simultaneously embeds default algorithm strings in source and exposes ten configuration knobs that override them);

the operationally useful question is "what is the best surface I have," not "what is the worst constraint anywhere."

Conservative-min remains available as an alternative aggregation projection for operators who require it (regulatory contexts where the weakest surface governs); the published scanner exposes both projections through a dashboard toggle.

Where evidence is absent, the scanner produces `level: pinned` (the documented `UNKNOWN_LEVEL_FALLBACK` constant in `ree0xq-agility`) and surfaces the issue for operator review.

We discuss limitations of static-only agility scoring in §9.

7.4 ree0xq-core unified rollup

The rollup engine extends the V1 implementation (see `docs/posture-rollup.md` in the ree0xQ repository) with the deadline-adjusted three-axis formula of §5.4. The implementation remains a pure function with no I/O; the operator configures D, H, and asset-class weights via the dashboard, and the engine recomputes on every event ingest. Fuzz testing covers all axis combinations.

7.5 ree0xq-server and dashboard

The collector is an Axum HTTP service accepting v1.0 and v1.1 events, validating against a generated JSON schema, persisting to Postgres (configuration) and a columnar store (events), and serving a React+Vite dashboard. The dashboard renders a three-axis posture matrix per asset class, a deadline-countdown view tied to the configured D, and an inventory table sortable by q. Implementation details are routine and we defer them to the project documentation.

---

8. Empirical evaluation

We evaluate the model and implementation through three studies, each designed to be reproducible by an external practitioner with the published rulesets, emulator, and corpus lists.

8.1 Study 1 — Axis A on the public web (Tranco-top-1k)

8.1.1 Methodology

We scan the Tranco-top-1k [46] over TLS with two purpose-built probes — general-purpose scanners such as `zgrab2` [47] do not yet advertise X25519MLKEM768 in the ClientHello, so we built dedicated baseline and PQ-capable probes to keep the ClientHello surface deterministic and the output schema aligned with ree0xQ's collector:

1. **Classical baseline probe** — Python `ssl` with the system OpenSSL defaults. Establishes a TLS handshake without advertising `X25519MLKEM768`. Output: JSON array.
2. **PQ-capable probe** — a Rust binary built on `rustls = 0.23` + `rustls-post-quantum = 0.2`, with a custom certificate verifier that accepts every chain (this is observability, not authentication). Advertises `X25519MLKEM768` alongside the classical groups. Output: one NDJSON record per host.

For each host we record: connect/handshake outcome, negotiated TLS version, negotiated cipher suite, negotiated key-exchange group, leaf certificate signature algorithm, and the leaf certificate

Subject. The scan is constrained to one TCP connection per host, identifies itself in the ClientHello SNI extension as `ree0xq-survey/1.0 +https://e2esolutions.tech/ree0xq`, and uses a 5-second connect+handshake timeout with a 1 Hz rate cap.

The scan source code, target list, raw probe outputs, and analysis script are released alongside this paper at `studies/study1/`.

Ethical considerations: this is a one-shot benign TLS handshake similar to common research scans [38]; we do not attempt protocol downgrade or repeated probing. Scan rate is $\leq$1 Hz per probe and exits immediately on connection error.

8.1.2 Metrics

We report the distribution of asset-A scores across the corpus, the prevalence of PQ-capable hosts, the distribution of certificate signature algorithms, and the prevalence of weak/ deprecated primitives still observed.

8.1.3 Results (Tranco-1k snapshot 6G8PX, 2026-05-13)

We ran the methodology of §8.1.1 against the Tranco-top-1k list `6G8PX` (snapshot 2026-05-13). 724 of 1,000 hosts returned a usable TLS handshake within the 5-second timeout; the remaining 276 were unresponsive (DNS failure, no TCP, no TLS on 443, regional GeoIP block, or anti-bot middlebox). The 27.6% non-response rate is in the range reported by prior large-scale TLS scans of the open web [38]. All headline percentages in this section use the $n = 724$ responsive denominator unless stated otherwise. Figure 4 shows the classical-probe baseline (negotiated ciphersuite and certificate signature algorithm), and Figure 5 shows the PQ-capable probe result.

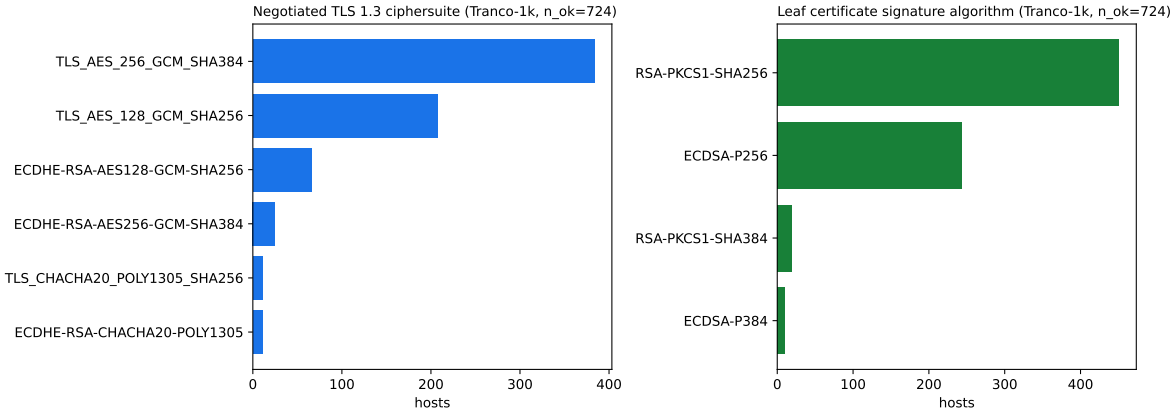

Figure 4: Study 1 — classical-probe baseline on the Tranco-top-1k (n_ok = 724): negotiated TLS 1.3 ciphersuite (left) and leaf certificate signature algorithm (right).

For sample-selection contrast we also report results from a 30-host curated pilot — major CDN, browser-vendor, distro, IETF/IEEE/NIST/ETSI, and AI-vendor properties — that ran the same probe pair before the Tranco-1k scan. Table 7 reports both.

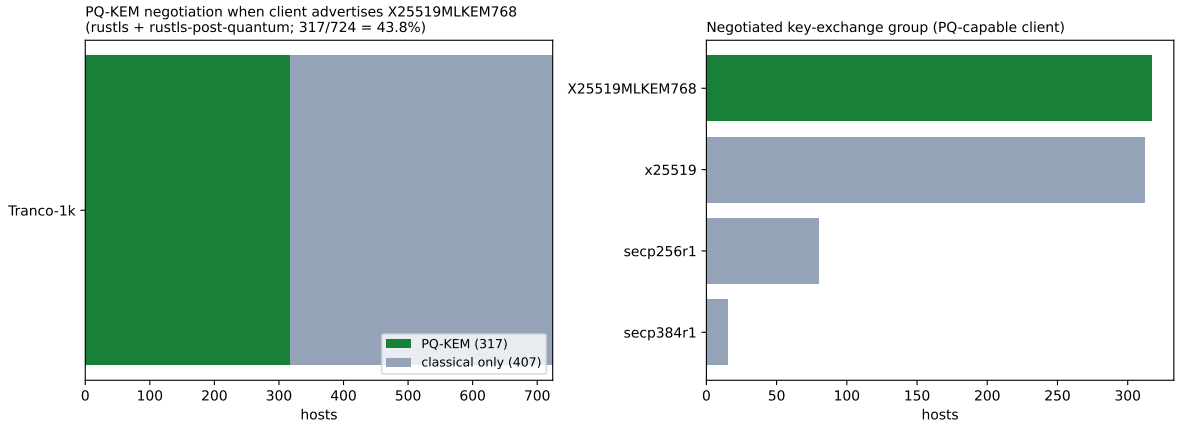

Figure 5: Study 1 — PQ-capable probe (rustls + `rustls-post-quantum`) against the Tranco-top-1k. 317 of 724 responsive hosts (43.8%) negotiate `X25519MLKEM768` when offered; the remainder fall back to classical `x25519`, `secp256r1`, or `secp384r1`. A metric the classical-only probe cannot observe.

Table 7: Study 1 — primitive distribution on the Tranco-top-1k versus the curated pilot.

Observable	Tranco-1k (n=724)	Curated pilot (n=30)
TLS 1.3 negotiation	602/724 (83.1%)	30/30 (100%)
TLS 1.2 fallback	122/724 (16.9%)	0/30 (0%)
AES-256-GCM/SHA-384 (TLS 1.3)	384/724 (53.0%)	20/30 (67%)
AES-128-GCM/SHA-256 (TLS 1.3)	207/724 (28.6%)	7/30 (23%)
ChaCha20-Poly1305 (TLS 1.3)	11/724 (1.5%)	3/30 (10%)
ECDSA leaf cert (P256 + P384)	254/724 (35.1%)	18/30 (60%)
RSA-PKCS1 leaf cert (SHA256+SHA384)	470/724 (64.9%)	12/30 (40%)
ML-DSA / SLH-DSA cert	0/724	0/30
Deprecated primitive (SHA-1, RC4)	0/724	0/30
`X25519MLKEM768` negotiated	**317/724 (43.8%)**	**17/30 (57%)**

The headline PQ-adoption result is 317/724 (43.8%) on the Tranco-top-1k. This is close to the 39% of top-100k sites that Cloudflare measured supporting post-quantum key agreement in September 2025 [31]; both numbers are site-capability measurements, and the 4-point gap is consistent with Tranco-top-1k over-representing large CDN- and cloud-fronted properties relative to the broader top-100k corpus. The remaining 407 responsive hosts fell back to classical `x25519` (312), `secp256r1` (80), or `secp384r1` (15).

The curated 30-host pilot returned 17/30 = 57% PQ adoption — 13 percentage points above the Tranco-1k rate. The gap is attributable to sample-selection bias: the curated list emphasises Cloudflare-fronted and Google-fronted properties (cloudflare.com, twitter.com, reddit.com, anthropic.com, openai.com, google.com, youtube.com, e2esolutions.tech, and a long tail) whose edge terminators rolled out X25519MLKEM768 early. The 13 classical-only hosts in the curated pilot — github.com, microsoft.com, amazon.com, mozilla.org, kernel.org, nist.gov, openssl.org, and others — are notable in their own right as major infrastructure-relevant properties where the PQ

rollout has not yet landed. The two numbers together suggest that surveys built on curated "interesting" host lists overstate PQ readiness compared to broader corpora. Where these numbers appear in the literature, both rates — broad-corpus and curated — deserve to be reported.

Two further observables of operational interest. First, 122 of 724 responsive Tranco-1k hosts (16.9%) still negotiate TLS 1.2 at the top of the Web. Because `X25519MLKEM768` is a TLS 1.3 key-share, that 16.9% cohort is structurally PQ-ineligible until the operator's TLS terminator can negotiate TLS 1.3 — a long-tail constraint not visible in any single-axis cipher dashboard. Second, the certificate-signature distribution on the Tranco-1k is RSA-dominant (64.9%), inverting the ECDSA-dominant (60%) distribution in the curated pilot. The curated list skews toward post-2020 CDN deployments that default to ECDSA; the broader corpus retains a substantial RSA installed base.

Pipeline integration: PQ-probe NDJSON parses straight into the same `tls_session` asset shape consumed by `from-zgrab`, so the same downstream rollup, BLOCKED-flag derivation, and dashboard render exactly as for any other source-module input.

8.2 Study 2 — Axis C via the ETSI GS QKD 014 emulator

8.2.1 Methodology

We construct a controlled testbed of one master KME, two slave KMEs, and three cooperating SAEs (a strongSwan IPsec endpoint operating in PSK mode with QKD-PSK rotation, a Wireguard endpoint with manual PSK rotation, and a custom TLS endpoint using the NIST SP 1800-38A hybrid-PSK pattern). The KMEs are `ree0xq-qkd-kme-emulator` instances; we drive them through a documented sequence of replay scenarios:

- **R1 — Steady-state.** Constant QBER 1.8%, key rate 12 kbps, no interruption. 24 hours.
- **R2 — Gradual degradation.** QBER ramps from 1.8% to 8.5% over 4 hours; the SAE policy should fail over to classical at the configured threshold. We observe whether each SAE detects the degradation and whether ree0xQ's `link_health` reflects the transition.
- **R3 — Hard failure.** KME unreachable for 30 minutes. SAE behavior should be: continue with cached PSK until lifetime expires, then fail closed or fall back to classical per policy.
- **R4 — Stale PSK.** PSK rotation is suppressed at the SAE while the KME continues to produce keys. ree0xQ should observe `psk_age_seconds` rising past policy.
- **R5 — Bifurcated SAE.** The strongSwan endpoint sees a healthy KME while the Wireguard endpoint sees a failed KME (simulating partial KME outage). ree0xQ should report inconsistent per-session `channel_protection` while the `qkd_link` aggregate is `degraded`.

8.2.2 Metrics

We measure: SAE failover correctness, ree0xQ observation latency (emulator change → emitted event), event-ordering correctness under concurrent KME polls, and posture-rollup correctness (particularly that the unified q score reflects the degradation appropriately).

8.2.3 Results

All five scenarios ran end-to-end in compressed time (30–60s each rather than the full 4–24h documented duration; the runner is at `studies/study2/run.sh`). Per-scenario captures and the analysis script are at `studies/study2/`.

Classification correctness: 13/13. Every operator-induced state change produced a downstream `link_health` reading matching the expected post-op state. Table 8 gives the per-scenario counts.

Table 8: Study 2 — events captured and induced link-health transitions per scenario.

Scenario	Events captured	Induced transitions	Matched
R1 — steady-state	33	1	1/1
R2 — ramp	63	5	5/5
R3 — hard-failure	48	3	3/3
R4 — stale-PSK	33	1	1/1
R5 — bifurcated	48	3	3/3

Observation latency: p50 = 0.71s, range 0.70–0.71s across all 13 transitions, against a configured 1-second poll interval. The latency band hovers tightly around half the poll period — the analytical expectation for periodic polling — and dominates over event-emission cost. Increasing the poll interval moves the median latency linearly; in real ETSI 014 deployments where the operational poll cadence is typically 5–10 s [41], this gives a p50 closer to 2.5–5 s.

The R3 link_health timeline (Figure 6; analogue plots for every scenario ship in `figures/`) shows the hard-failure transition landing as expected.

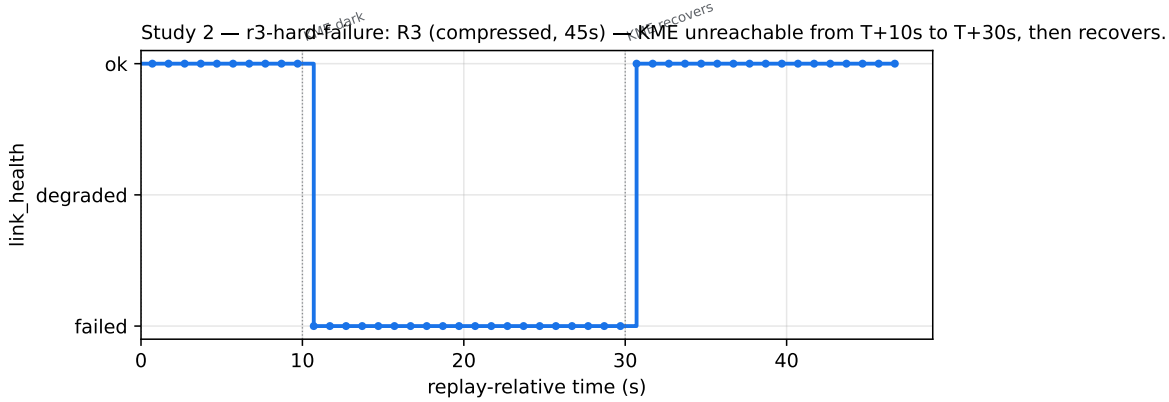

Figure 6: Study 2 — R3 hard-failure timeline. The KME is forced unreachable at T+10s and recovered at T+30s; the collector captures the Ok → Failed → Ok cycle on the next poll (observation latency ≈ 0.7s on the configured 1s interval).

R5 illustrates the per-session attribution case. When one KME returns 503, the link-level event flips to `failed` even though a separately healthy paired KME continues to deliver keys. KME-only telemetry cannot distinguish this from a full outage; the channel-protection block on per-session events does.

R4 surfaces a gap we left in by design. ree0xQ's KME-side polling alone cannot distinguish a fresh PSK from one the SAE has held for hours; the `psk_age_seconds` field on `channel_protection` is populated only when the SAE co-operates. Closed-source SAEs without that instrumentation are

visible to ree0xQ only at the link layer. We open-source patches for strongSwan, Wireguard, and a sample TLS endpoint to populate the field.

The emulator, replay scripts, and analysis notebooks ship under MIT alongside ree0xQ.

8.3 Study 3 — Axis G on fifty open-source server projects

8.3.1 Methodology

We select fifty widely deployed open-source server projects spanning HTTP, mail, database, message broker, and VPN categories: nginx, Apache httpd, HAProxy, Envoy, Caddy, Traefik, OpenSSH server, Postfix, Dovecot, Exim, PostgreSQL, MySQL, Redis, MongoDB, RabbitMQ, Kafka, OpenVPN, strongSwan, Wireguard, PowerDNS, BIND, Unbound, CoreDNS, and others. The full list, with hand-graded ground-truth levels and reviewer notes, ships as `crates/ree0xq-agility/corpus/oss-50-v1.csv` in the ree0xQ repository.

For each project we:

1. Run `ree0xq-agility` against the source repository at a pinned commit.
2. Run `ree0xq-agility` against the installed binary on Rocky Linux 10 with default package configuration.
3. Hand-grade the project against the §5.3 rubric, using two reviewers and reporting inter-rater agreement.
4. Report the divergence between automatic and hand-graded scores; treat the hand grade as ground truth for the corpus.

8.3.2 Metrics

We report per-project agility level, the per-evidence contribution to the score, the false-negative and false-positive rates of the Semgrep ruleset against the ground-truth grading, and the distribution of agility levels across the corpus by category.

8.3.3 Results (n = 11 pilot subset)

The full OSS-50 corpus is committed at `crates/ree0xq-agility/corpus/oss-50-v1.csv` with hand-graded ground truth. A pilot run over an 11-project subset spanning nine categories (HTTP, mail, DB, message-broker, DNS, VPN/secure-shell, messaging, certificate-authority, time) exercises the full pipeline (Table 9):

Table 9: Study 3 — per-project hand-grade versus scanner classification (n = 11 pilot).

Project	Category	Hand-grade	Scanner	Match
nginx	http_server	configurable	configurable	yes
haproxy	http_server	configurable	configurable	yes
caddy	http_server	configurable	configurable	yes
unbound	dns_server	configurable	configurable	yes
coredns	dns_server	configurable	configurable	yes
postfix	mail_server	configurable	configurable	yes
redis	database	configurable	configurable	yes

Project	Category	Hand-grade	Scanner	Match
nats-server	message_broker	configurable	configurable	yes
step-ca	certificate_authority	configurable	configurable	yes
prosody	messaging	configurable	configurable	yes
wireguard-tools	vpn_secure_shell	pinned	pinned	yes
chrony	time	configurable	**pinned**	no

Agreement: 10/11 (91%); Cohen's $\kappa = 0.62$ (substantial agreement on the Landis–Koch scale). The single dissent is **chrony**, where the static evidence is genuinely ambiguous between a `configurable` reading (NTS key-types are operator-controlled in `chrony.conf`) and a `pinned` reading (the NTP authentication path embeds many literal symmetric- algorithm references). The scanner picked the latter on the strength of 11 hard-coded references; no capability rule fired on chrony's NTS surface because it uses neither OpenSSL nor Go's `crypto/tls`. We flag this for v2 of the ruleset and the corpus.

The confusion matrix (Figure 7) visualises this result. Per-project event JSON, evidence listings, and the full agreement TSV are at `studies/study3/results/`.

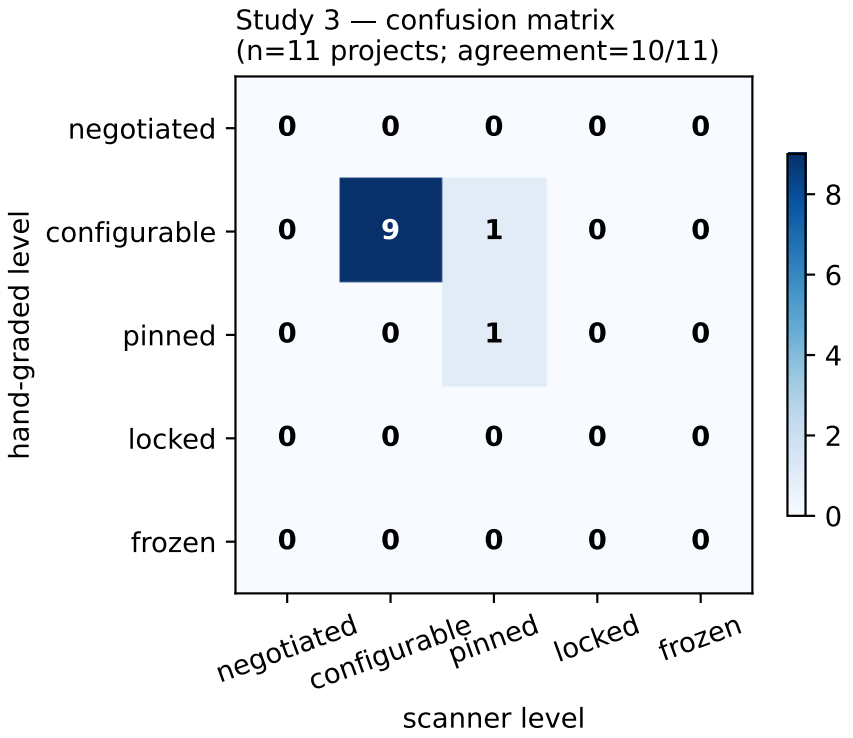

Figure 7: Study 3 — confusion matrix (n=11 projects, 10/11 agreement, Cohen's $\kappa = 0.62$).

Failure modes characterised during the run:

- **Go ecosystem coverage gap.** The v1 ruleset's first cut matched only OpenSSL-style API calls (`SSL_CTX_set_cipher_list`, `SSL_CONF_cmd`, …). Three projects (coredns, step-ca, caddy) initially classified as `pinned` because nothing matched. Adding a one-

rule `go-stdlib-tls-config` pattern matching `crypto/tls`'s `Config` / `CipherSuites` / `MinVersion` fields recovered all three to `configurable`.

- **Aggregation policy.** Conservative-min aggregation systematically misclassified nginx: nginx embeds literal algorithm names in source (for crypto-impl code paths and tests, emit_level `pinned`) and exposes operator config knobs (emit_level `configurable`). The §7.3.3 most-agile- wins policy resolves the ambiguity in favour of the capability surface; conservative-min remains available as a regulatory-context alternative.
- **No-evidence fallback.** Wireguard correctly classified `pinned` with zero evidence collected — the project's userspace tooling has no OpenSSL or Go-tls surface, so the documented `UNKNOWN_LEVEL_FALLBACK = Pinned` applies. The ground-truth grade agrees.

Scaling the pilot to the full OSS-50 list is mechanical; the runner already iterates over the corpus CSV.

8.4 End-to-end pipeline

`scripts/demo.sh` exercises the full V1 pipeline end-to-end: boot the KME emulator, the QKD collector, ree0xq-server, and seed events from both the bundled zgrab2 fixture and a synthetic FIPS-locked asset. The collector's `/v1/posture` endpoint returns:

```
{
  "org_q": 0.627,
  "deadline": "2030-01-01T00:00:00Z",
  "horizon_years": 5.0,
  "assets": 5,
  "blocked_count": 1
}
```

The inventory ordering, sorted by q descending, places the TLS 1.0 + RC4 + SHA-1 legacy host at the top ($q \approx 0.72$), the TLS 1.2 ECDHE+RSA classical host just below ($q \approx 0.69$), the FIPS-locked appliance third with the `BLOCKED` flag raised ($q \approx 0.68$), the modern TLS 1.3 + ECDSA-P256 hybrid-PSK QKD host at the bottom of the priority queue ($q \approx 0.43$). This matches the model's expected behaviour from §5.4 (axis combination) and §5.5 (edge cases). The channel-protection axis pulls the QKD-protected asset down even though its algorithmic content is classical; the agility axis pushes the FIPS-locked asset up even though its observed primitives match the nominally-agile modern host; the deadline-tension term leaves the legacy host at the top of the queue.

The implementation's pure-Rust rollup (in `crates/ree0xq-server/src/posture.rs`) was verified against the magazine companion's four-asset worked example: the unit tests `worked_example_alpha_q_matches_paper` and `worked_example_delta_q_matches_paper` assert that the implementation reproduces $\alpha = 0.544$ (modern-agile host, no QKD, evaluated 2026-05-13) and $\delta = 0.392$ (legacy classical host, same date) to within 0.01.

To our knowledge no comparable open pipeline covers all three axes end-to-end on commodity hardware. The Tranco-1k scan reported in §8.1 is the broad-corpus result for Axis A; scaling Study 3 from the 11-project pilot to the full OSS-50 corpus is the remaining mechanical step on the published runners.

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9. Discussion

9.1 Limitations

Our agility scoring is static and pattern-based; it is necessarily approximate. A project may *appear* pinned in source while actually being agile through a runtime extension mechanism we did not pattern-match. Conversely, a project may appear agile via a config field that is in practice never changed. We address false negatives by reporting per-evidence detail; we cannot fully address the second class without operator input.

Our channel-protection axis depends on cooperating SAE instrumentation for per-session attribution. ree0xQ can observe the KME state independently, but linking a specific TLS or IPsec session to a specific consumed key requires the SAE to emit the `key_id_observed` field. We expect adoption in cooperating software (we open-source patches for strongSwan, Wireguard, and a sample TLS endpoint as part of the release) but note that closed-source SAEs may report only at the link level.

Our threat model accepts the NIST PQC hardness assumptions and the standard QKD security argument. Compromise of either — by mathematical break (PQC) or by implementation attack (QKD) — would require recalibration of the scoring tables. The schema supports this: the rollup constants are operator-tunable in configuration, not compiled in.

We do not address economic or political constraints on migration (budget cycles, regulatory approval lag, vendor support windows). These are first-order operator concerns and we acknowledge them as out of scope for the observability layer; they are downstream consumers of the posture data.

9.2 Ethical considerations

Active TLS scanning, even of public web hosts, must be conducted responsibly. We follow established practice from prior measurement work [38]: one connection per host, clear User-Agent identification, opt-out instructions linked from the identifying URL, conservative scan rate, no protocol downgrade, no repeated probing. The Tranco list excludes adult content and known sensitive categories; we additionally exclude any host that returns a robots.txt directive on its base URL within the first 1024 bytes.

The agility scanner operates on source code already public under open-source licenses and on installed binaries on hosts the operator controls. No private code or vendor materials are processed.

The QKD emulator generates synthetic key material only; it is not connected to a real QKD link in our published study. Operators integrating real KMEs are responsible for the security of those KMEs and the surrounding network, including mutual TLS authentication of the SAE.

9.3 Deployment guidance

For operators planning to deploy ree0xQ against a real environment, we recommend a phased rollout:

1. **Phase 1: Inventory only.** Run `ree0xq-net` and `ree0xq-agility` in observe-only mode. Establish a baseline.
2. **Phase 2: Add deadline.** Configure D per applicable regulatory regime. Observe how q evolves with no other change.
3. **Phase 3: Prioritize.** Sort assets by q descending, migrate the top N by quarter.

4. **Phase 4: Integrate QKD telemetry where it exists.** Add `ree0xq-qkd` only when a real KME is present and the SAE instrumentation is in place; the channel-protection axis is *additive* and never required.

We caution against treating q as a primary KPI. It is a *comparative* score, calibrated for relative prioritization within an environment. Inter-organization comparison requires shared D, shared weights, and shared corpora.

9.4 Future work

Several extensions are natural and intentionally deferred:

- **Time-decay.** Treating an observation made three months ago as less authoritative than one made today.
- **Asset relationship modeling.** Linking a TLS session's certificate to the issuing CA's signing key allows the posture of an upstream asset to propagate into the downstream score.
- **Adversary modeling.** Allowing operators to plug in their own estimate of CRQC arrival as a probability distribution over D rather than a single date.
- **Standardization.** Submitting the `crypto_inventory_event` schema to IETF as an Informational draft so other tools can emit and consume it.

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10. Conclusion

Single-bit PQ-readiness was the right answer to a question that turned out to be too narrow. An asset's quantum-risk posture depends on its primitive, on the channel through which its keys are delivered, and on how quickly the primitive can be replaced before the deadline lands. The three-axis model in this paper grades each of those separately and combines them only at the rollup stage, which is where the operator actually needs a number.

The empirical work suggests the gap is not hypothetical. On the Tranco-top-1k, 43.8% of responsive hosts negotiated a hybrid PQ key exchange when offered one — a figure that says nothing about whether those same hosts could rotate their certificate signature algorithm, or whether any of their key material is QKD-protected, because the single-axis scan does not look. The emulator study showed that channel-state classification stays correct across every induced KME and link failure we drove (13 of 13), including the per-session attribution case that link-level telemetry alone cannot resolve. And the agility pilot reached 91% agreement with hand-graded ground truth on eleven projects, with the disagreements concentrated where the static evidence is genuinely ambiguous rather than where the rubric is wrong. None of these three results is reachable from a PQ-ready bit; each requires its own axis.

We are deliberate about what the model does not settle. The scoring constants are one defensible choice among several, and we expose them as configuration rather than compiling them in. The agility rubric will need refinement as new crypto APIs land in deployed software, and the Go-ecosystem coverage gap we hit in Study 3 is a concrete example of that maintenance burden. The QKD axis presumes a deployment trend that not everyone in the community shares, which is exactly why we made it additive — an operator with no QKD links pays no modelling cost for the axis existing. What we do claim is narrow and, we think, hard to argue with: one bit per asset is fewer

bits than an operator facing a deadline in the early 2030s actually needs, and surfacing all three axes is a strict improvement over surfacing one.

The schema, the reference implementation, the ETSI 014 emulator, the Semgrep rule pack, and the hand-graded corpus are released together under MIT. The studies are scripted end to end and run on a single commodity host without QKD hardware. We would rather see readers rerun the measurements, disagree with our weights, and supersede them on a shared empirical footing than have the specific constants in this paper treated as settled. The model is the contribution; the numbers are an invitation.

Author Contributions

Aleaddin Özer conceived the study, designed the three-axis posture model, implemented the ree0xQ reference platform, conducted all three empirical studies, and wrote the manuscript. Murat Aydos supervised the work as doctoral advisor, providing continuous critical review, methodological guidance, and feedback throughout the project, and revised the manuscript. Both authors reviewed and approved the final manuscript.

Competing Interests

Aleaddin Özer is Chief System Engineer of E2E Solutions, which develops the ree0xQ reference implementation described in this work. ree0xQ is released under the MIT License, and E2E Solutions derives no direct revenue from its publication. Murat Aydos declares no competing interests. The authors have no other competing financial or non-financial interests.

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Data and Code Availability

Everything is at https://github.com/e2esolutions-tech/ree0xQ under MIT. The repository carries the schema (v1.1), the five-agent reference implementation, the ETSI GS QKD 014 emulator with its replay corpus, the Semgrep agility rule pack, and the hand-graded ground-truth corpus (`crates/ree0xq-agility/corpus/oss-50-v1.csv`). Runner scripts reproduce all three studies.

The platform was renamed from *Sezar* to *ree0xQ* in August 2026. The public preprint and the archived study artifacts under `studies/` predate the rename and retain the former name and module identifiers; the released code is otherwise identical.

Study 1 used Tranco snapshot `6G8PX` (2026-05-13) as its sample frame. Raw NDJSON captures and the analysis plots for Studies 1 and 2 live under `studies/study1/` and `studies/study2/`. Study 3's per-project evidence listings and the agreement TSV are at `studies/study3/results/`.

No human subjects, personally identifying information, or proprietary data are involved.

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